Cortical Specifi cation and Neuronal Migration 19 how it relates to the neocortical growth of higher mammalian species. Several studies have noted that brain size is correlated with an enlargement of both subdivisions of the SVZ and with increased numbers of IPCs. In particular, the gyrifi cation index of a wide array of species has been linked to the size of the bRGC population (Fietz et al. 2010; Hansen et al. 2010). While species with a lissencephalic cortex contain bRGCs, and some (e.g., marmoset) even develop a distinct oSVZ compartment (Garcia-Moreno et al. 2012; Kelava et al. 2012), the numbers of bRGCs within these brains are reduced relative to gyrenceph- alic species. Nonetheless, how bRGCs contribute on a macro scale to cortical formation is still at a theoretical stage. For instance, the human brain has ex- panded more in the lateral dimension than in the radial dimension (thickness) and it is unclear how and whether bRGCs lead to a preferential increase surface area rather than thickness. Because of these unanswered questions, a primary mechanism driving species-specifi c differences in brain growth remains the expansion of the founder cell population prior to the onset of neurogenesis (Figure 2.1). Taken together, differences in brain size across the phylogenetic tree are likely due to the synergistic effects of a larger founder cell population and species-specifi c differences in the composition of the IPC pool as well as changes in the total neuronal output per precursor type. A second role for the precursor heterogeneity in brain development may be that it contributes directly to neuronal diversity. The classes of excitatory neurons which comprise the six layers of the neocortex have been character- ized by their birth date, molecular expression, electrophysiology, and target- specifi c projections. Clonal analysis by genetic fate-mapping has shown that individual aRGCs can produce neurons that span all of the neocortical lay- ers, suggesting that aRGCs may be progressively tuned over developmental time to generate different types of excitatory neurons. Transplantation experi- ments indicate that isolated aRGCs generate neuron types appropriate for their birth date, even when placed in a heterochronic environment (McConnell and Kaznowski 1991), further supporting the notion that aRGCs undergo temporal fate restriction. Another assumption contained within this progressive fate re- striction model is that aRGCs of any given developmental age are all identical, but in vivo evidence for this assumption is thus far unconvincing. It is also important to note that the molecular mechanisms underlying this temporal fate restriction have not yet been identifi ed. Nevertheless, while this model may explain a primary mechanism for producing the different neuronal cell types across neocortical layers, it does not describe a method for generating different types of neurons within each layer. Morphological and electrophysiological studies have defi ned distinct intra- laminar populations of excitatory cells and shown that intralaminar diversity differs across areas. At any given time during the neurogenic interval, neu- rons are contemporaneously produced directly from aRGCs as well as indi- rectly from multiple classes of IPCs. As shown from birth-dating studies, these neurons are destined for the same neocortical layer, and mouse studies have From "The Neocortex," edited by W. Singer, T. J. Sejnowski and P. Rakic. Strüngmann Forum Reports, vol. 27, J. R. Lupp, series editor. Cambridge, MA: MIT Press. ISBN 978-0-262-04324-3